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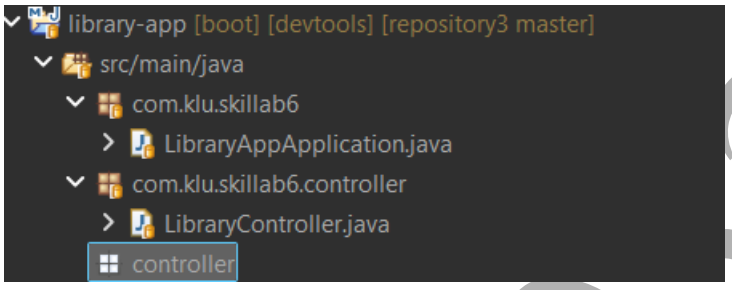
SECTION:	S-8
SUBJECT:	FULL STACK APPLICATION DEVELOPMENT

#SKILL - 6 Spring MVC Web Request Handling Demo

Tasks:

1. **Create a controller class named LibraryController.**

   <u>Output:-</u>



2. **Create a method mapped to /welcome that returns a welcome message.**

   <u>Output:-</u>

```java
package com.klu.skillab6.controller;

import java.awt.print.Book;

@RestController
public class LibraryController {


List<Book> bookList = new ArrayList<>();

// 1. Welcome message
@GetMapping("/welcome")
public String welcome() {
    return "Welcome to the Library System";
}

```

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3. **Create a method mapped to /count that returns an integer representing total books.**

Output:-

```
// 2. Total books count
@GetMapping("/count")
public int count() {
    return 100;
}
```

4. **Create a method mapped to /price that returns a sample book price as double.**

Output:-

```
// 3. Book price
@GetMapping("/price")
public double price() {
    return 499.99;
}
```

5. **Create a method mapped to /books using @GetMapping that returns a list of book titles.**

Output:-

```
// 4. List of book titles
@GetMapping("/books")
public List<String> books() {
    List<String> titles = new ArrayList<>();
    titles.add("Spring Boot Guide");
    titles.add("Java Programming");
    titles.add("Microservices with Spring");
    return titles;
}
```

6. **Create a method mapped to /books/{id} using @GetMapping and return book details using @PathVariable.**

Output:-

```
// 5. Book by ID
@GetMapping("/books/{id}")
public String getBookById(@PathVariable int id) {
    return "Book details for ID: " + id;
}
```

7. **Create a method mapped to /search using @GetMapping that accepts a request parameter (title) and returns a confirmation message.**

Output:-

```
// 6. Search book by title
@GetMapping("/search")
public String searchBook(@RequestParam String title) {
    return "Searching for book: " + title;
}
```

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8. Create a method mapped to /author/{name} using @GetMapping that returns a formatted message with the author's name.

Output:-

```
// 7. Author name
@GetMapping("/author/{name}")
public String author(@PathVariable String name) {
    return "Books written by author: " + name;
}
```

9. Create a method mapped to /addbook using @PostMapping that accepts a Book object from the request body and adds it to an in-memory list.

Output:-

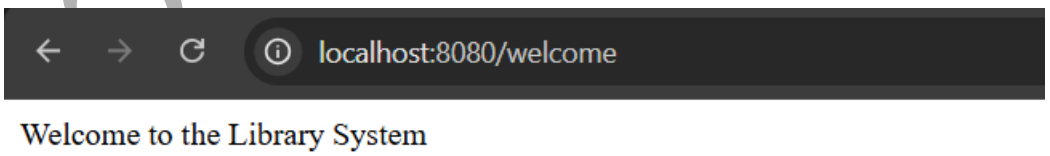
```
// 8. Add new book
@PostMapping("/addbook")
public String addBook(@RequestBody Book book) {
    bookList.add(book);
    return "Book added successfully";
}
```

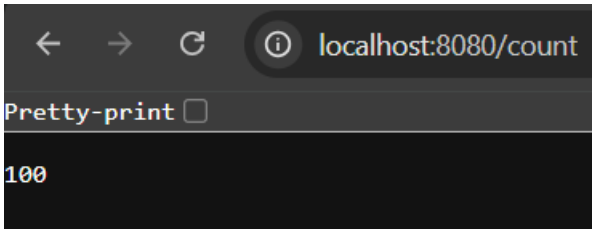
10. Create a method mapped to /viewbooks using @GetMapping that returns all added Book objects.

Output:-

```
// 9. View all books
@GetMapping("/viewbooks")
public List<Book> viewBooks() {
    return bookList;
}
```

Final Outputs:-





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localhost:8080/price

Pretty-print

```
499.99
```

localhost:8080/books

Pretty-print

```
["Spring Boot Guide","Java Programming","Microservices with Spring"]
```

localhost:8080/books/1

Book details for ID: 1

localhost:8080/search?title=Spring

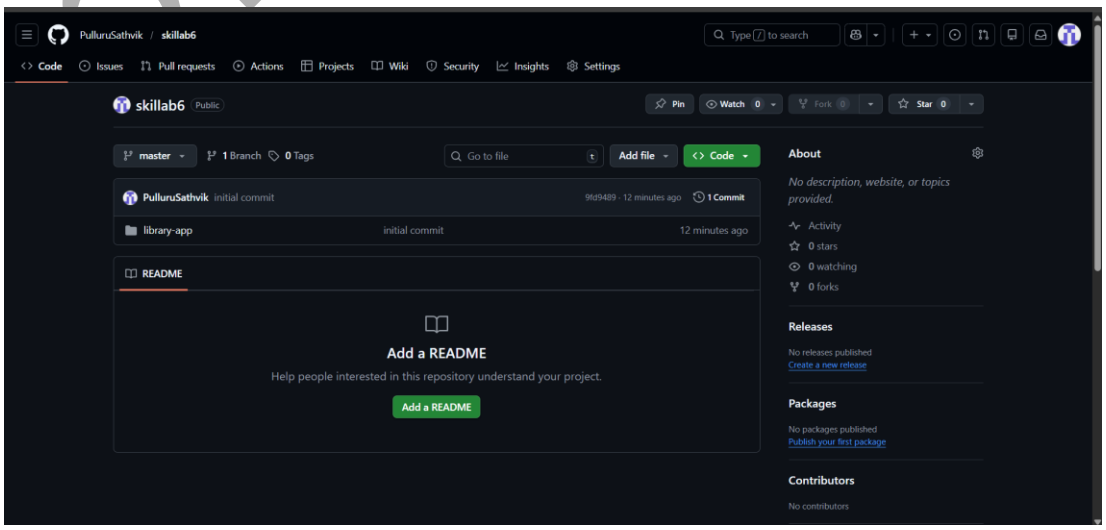
Searching for book: Spring

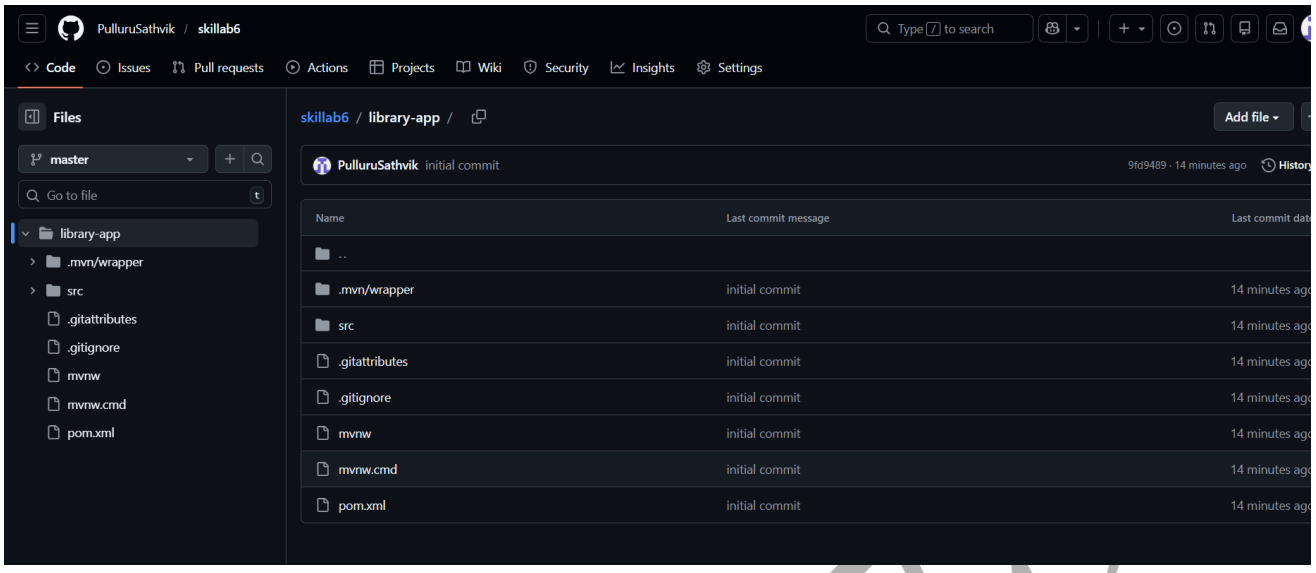
localhost:8080/author/James

Books written by author: James

11. Push the Spring Boot project to GitHub.

Output:-





GitHub Link:- https://github.com/PulluruSathvik/skillab6/tree/master/library-app

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